Project Overview: Key Features

Interactive UI

Built with JavaFX to deliver a smooth and engaging visual experience for players.

Film Guessing Core

The central mechanic involves players predicting film names from provided actor clues.

Database Integration

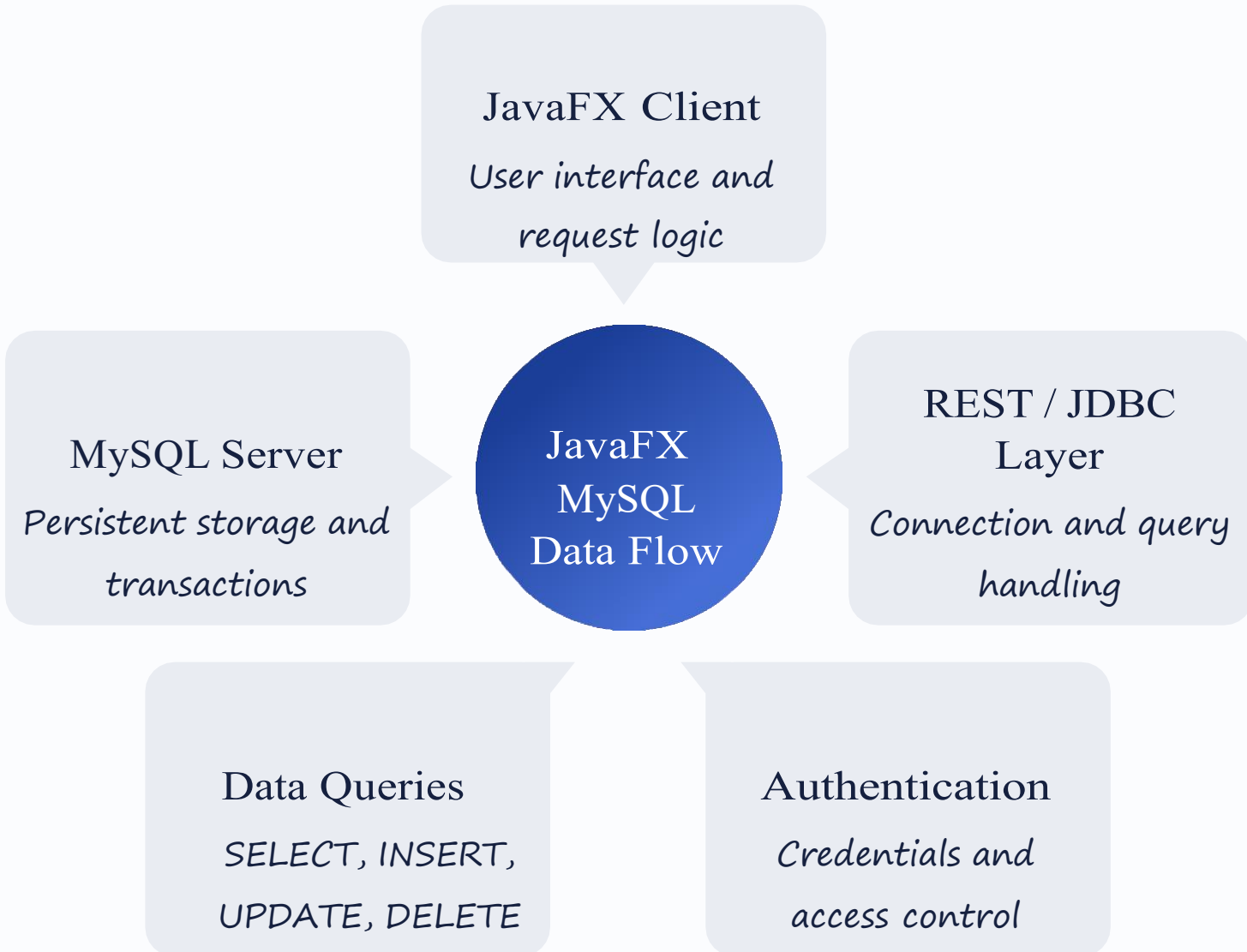
Efficient data management using MySQL for film data, player progress, and in-game economy.

Game Progression

Features a scoring mechanism (coins) and defined level progression to track user achievement.

System Architecture: Client-Server Model

The application adopts a client-server architectural principle, with the JavaFX application serving as the client front-end and MySQL acting as the data backend.



Project Structure

- **FilmGame:** Contains the core JavaFX UI components and primary game logic.

- **database:** Dedicated package for all MySQL database interaction methods and connection management.
- **images:** Repository for graphical assets used throughout the user interface.
- **sound:** Stores audio files for background music and in-game sound effects.

Core Technology Stack

A focused and robust technology stack ensures performance and maintainability across the application.

JavaF	Dedicated framework for responsive and interactive User Interface (UI)
MySQ	Relational Database Management System (RDBMS) for efficient and structured data
Jav	The core programming language governing all application logic and

JavaFX User Interface Components

The **FilmGame** package defines the structure and behaviour of the user-facing application screens.

1	Main.java The application's entry point, extending `javafx.application.Application` to initialise and launch the primary interface (MainFrame).
2	MainFrame.java Likely serves as the initial screen or main menu, handling application flow and navigation.
3	LevelsFrame.java Manages the display and selection of available game levels, showing player progress.
4	PlayFrame.java The central game screen where users actively interact with the film guessing mechanism, input answers, and utilise hints.
5	MusicAndSound.java Dedicated class to manage and control background music and in-game sound effects for enhanced immersion.

Database Integration via ManageDataBase

*The **database** package contains crucial classes that abstract the complexity of data persistence and retrieval from the MySQL server.*

DataBaseConnection.java

Responsible for establishing the critical connection to the MySQL database instance. It uses the JDBC driver with the connection string:

```
jdbc:mysql://localhost/films?
user=root&password=******
```

Note: This connection string must be configured with specific credentials during setup.

Setting Up the Game Environment

Successful deployment requires several essential components and adherence to specific setup steps.

JDK Installation

Ensure you have the Java Development Kit (JDK) version 8 or higher installed to support JavaFX.

MySQL Server

A running instance of the MySQL Server is necessary for data storage and retrieval.

MySQL JDBC Driver

This driver must be included in your project's classpath to enable Java-to-MySQL connectivity.

Installation Steps: Database and Codebase

Follow these steps to clone the repository, configure the database, and prepare the application for running.

1

Clone the Repository

```
git clone
https://github.com/SeifEldenOsama/film_name_Expectation
```

2

Database Schema Creation

Create the MySQL database named **films**. *Define the necessary tables:* film_actors, game_state, *and* coins.

3

Populate Data

Fill the film_actors *table with film titles and corresponding actor names. Set initial values for the* game_state *and* coins *tables.*

4

Update Connection

Modify the database connection string in

ith your

`DataBaseConnection.java` dentials (username and password).

Build and Run

The final phase involves compiling the source code and executing the JavaFX application.

Compile with IDE

Use a Java IDE (e.g., IntelliJ IDEA or Eclipse) configured for JavaFX projects to compile the

Include JDBC Driver

Crucially, ensure the MySQL JDBC driver JAR file is correctly added to your

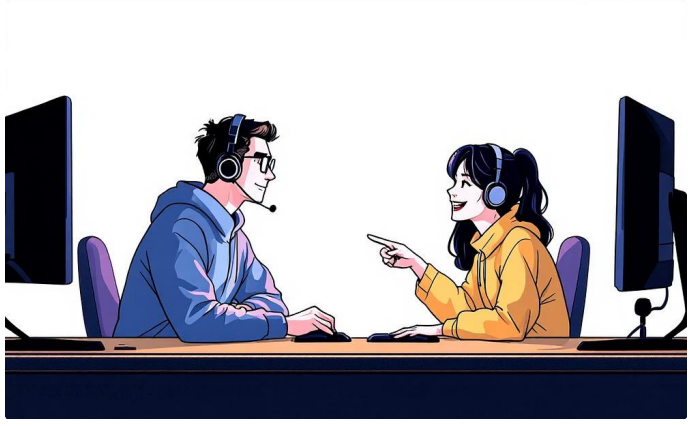
Execute Application

Run the compiled application via the IDE or using standard Java execution commands.

> **Note:** Correct configuration of the JDBC driver is critical for preventing runtime database connection errors.

Future Directions and Potential Enhancements

Based on the project's foundation, several compelling features are slated for future development to expand game complexity and user interaction.



Advanced Hints System

Implementing more sophisticated and contextual hints beyond simple actor names to aid players.

Sophisticated Scoring

Introducing detailed scoring mechanisms, score multipliers, or leaderboards for

competitive play.

Multiplayer Support

Integrating online functionality to allow multiple users to play simultaneously or compete head-to-head.